

Chenxi Lin

cl2732@cornell.edu • (917) 225-1626 • New York, NY • [linkedin.com/in/chenxilin4312](https://www.linkedin.com/in/chenxilin4312)

EDUCATION

Cornell University, College of Engineering

Bachelor of Science in Biomedical Engineering

Ithaca, NY

May 2028

PROFESSIONAL EXPERIENCE

Joey ELM, Founder

June 2026 – Present

Ithaca, NY

- Built joey ELM, a bootstrapped AI research assistant for wet labs, and shipped it to production alone in 30 days (191 commits, ~27K lines of TypeScript on Postgres and Supabase), now used by 10+ student researchers at Cornell.
- Gave the agent 16 tools on a Claude tool-use loop with write access to notebook entries, reagent counts, equipment bookings, and to-dos, replacing five separate systems with plain-language requests.
- Wired agent answers to live, lab-scoped searches of OpenAlex, Semantic Scholar, and the researcher's own notebook, which cut literature reviews that took hours down to seconds.

Lee Lab, Cornell University, Research Assistant

May 2025 – Present

Ithaca, NY

- Prepare 300+ NGM plates weekly, including custom auxin and RNAi plates, for high-throughput *C. elegans* lifespan and stress-resistance studies.
- Train and manage two undergraduates on plate production and maintain general lab stocks including LB media, antibiotics, and molecular biology buffers.

Roeder Lab, Cornell University, Undergraduate Researcher

December 2024 – Present

Ithaca, NY

- Built a MATLAB biomechanical model of plant-to-animal cell transformation with tunable cell wall thickness, shape, and osmolyte concentration, then used it to show that geometry drives growth behavior while thickness and osmolyte concentration only scale it.
- Validated the model against multiple large experimental datasets with curve fitting and stochastic modeling, which improved its predictions of stress-strain and osmoregulatory behavior.
- Optimized protoplast isolation and growth protocols and tracked morphology in real time by live-cell confocal microscopy to validate the model, in collaboration with the Odde Lab (Minnesota) and Boudaoud Lab (École Polytechnique) under the NSF URoL:EN grant.
- Running laser ablation experiments on Arabidopsis sepals to test whether releasing tissue tension causes the surrounding epidermis to buckle, imaging the ablated region by confocal microscopy and quantifying cell shape change through segmentation.
- Generating the imaging and segmentation dataset that a collaborating lab is using to build a computational model of buckling.

Blenis Lab, Weill Cornell Medicine, Undergraduate Researcher

July 2022 – August 2024

New York, NY

- Led an independent project showing elevated BCAA levels increased prostate cancer metastasis, and demonstrated experimentally that low-dose TGF- β induces EMT in lung cancer.
- Utilized a wide range of laboratory techniques, including proteomics, metabolomics, gel electrophoresis, 3D and 2D tissue culture, and CRISPR.

EXTRACURRICULAR EXPERIENCE

Cornell Maker Club, Member

January 2026 – Present

Ithaca, NY

- Use CAD, laser cutting, and FDM printing for personal projects and help other members troubleshoot equipment.

Cornell Lambda Phi Epsilon, Vice-President External

March 2025 – Present

Ithaca, NY

- Coordinate social events and collaborations with other social and cultural organizations on campus.
- Lead fundraising efforts as Fundraising Chair, raising over \$2,000 per semester.

Cornell Undergraduate Research Board, Student Mentor

March 2024 – Present

Ithaca, NY

- Mentored cohorts of 6–10 undergraduate students, providing guidance on research involvement within the field.

HONORS AND AWARDS

Cornell Summer Experience Grant

June 2025

- Awarded a \$5,000 competitive Cornell Summer Experience Grant to support summer research.

APIA/United Health Foundation Scholarship

May 2024

- Awarded a \$10,000 scholarship in recognition as part of the Diverse Scholars Initiative.

SKILLS

MATLAB, Python, R, SQL, Ansys, MorphoGraphX, Fiji/ImageJ, Mandarin (Native)